

Data visualization checklist

First impressions

1. Can someone look at this and understand the main message right away?
2. Is there a meaningful title that supports the main message?¹
3. Is the type of visualization appropriate for the data being displayed?²
4. Does it take up too much or too little space on the page? Is it centered?
5. Have all coded variable names been replaced with text that is easy to understand?
6. Does it look professional enough to be included in a published journal article, newspaper, or blog?³

Graphs

7. Are both the x-axis and y-axis labeled with the units clearly specified?⁴
8. Have appropriate intervals been selected for the tick labels on each axis?⁵
9. Does the size of the text or numbers need to be increased or decreased?⁶
10. Do any decimal places need to be added or removed?
11. Is it necessary to include a legend?⁷
12. Is there too much information plotted so the graph is difficult to read and interpret?
13. Is an appropriate thickness used to display the bars, lines, dots, etc?⁸
14. Would the graph look better by changing the colors?⁹

Tables

15. Does each column have a meaningful name that does not result in the column being too wide?
16. Are there too many columns?
17. Do any decimal places need to be added or removed?

¹In addition to the title, in many cases it's useful to include a subtitle that states a key takeaway, provides additional information, or specifies the units (e.g. billions of dollars).

²Is a table or graph more appropriate? For graphs, note that certain types are more useful than others depending on the intended purpose. For example, to show correlation use a scatter plot, to display data over time use a line chart, to compare static values use a bar chart, to plot the distribution use a histogram or density plot, to show composition use a pie chart, etc.

³When making tables in R, use the `kableExtra` package. When making graphs in R, use `ggplot`. The visual presentation of `ggplot` graphs can often be improved by using the `ggplotly` command from the `plotly` package.

⁴If a graph's units are already specified in the title or subtitle, it's not necessary to include the units again in the y-axis label. If the x-axis of a graph displays a series of dates, it's not necessary to assign the label "Date" because it's already clear to the reader that dates are being shown.

⁵Nice intervals are preferred to less natural intervals. For example, when displaying a series of dates, it's better to use the labels 1990, 2000, 2010, 2020 rather than say 1991, 1998, 2005, 2012, 2019.

⁶In many graphs the size of the text or numbers is too small. A good rule of thumb is to always increase the text size unless doing so would make the text too large.

⁷For graphs that require a legend, moving the legend to the right of the graph, beneath the graph, or directly on the graph can make things look neater. Note that it's often unnecessary to include a legend title. For example, if a legend lists different countries, it's not necessary to assign the label "Country" because the reader already knows these are countries.

⁸The default size R assigns to plotted lines and dots in `ggplot` is often too small and needs to be increased.

⁹The default R color scheme in `ggplot` is somewhat odd and custom colors almost always look better. It's important to ensure there is enough distinction between different colors that are used so the reader can easily distinguish one series from another. In the case of line graphs, using a mix of dashed, dotted, and solid lines can be a helpful way to more clearly identify different series.